

Clinic Appointment System: Class Design Document

1. Architectural Overview

The system is built using Object-Oriented Programming (OOP) principles, specifically utilizing **Inheritance** and **Polymorphism**. The architecture features a base User class to handle shared authentication and identification attributes, which branches into three specialized child classes: Admin, Doctor, and Patient. A standalone Appointment class acts as the relational bridge between Doctors and Patients.

All classes feature a polymorphic to_dict() method to ensure smooth serialization for JSON file storage.

2. Base Class: User

Description: The parent class for all human entities in the system. It encapsulates the core data needed for authentication and basic contact.

Attribute	Data Type	Description
user_id	String	Unique identifier (e.g., "A-100", "DR-201", "P-101").
name	String	Full name of the user.
pin	String	4-digit security PIN for system login.
phone_number	String	Contact phone number.

Method	Parameters	Return Type	Description
__init__()	user_id, name, pin, phone_number	None	Constructor to instantiate a new User.
verify_pin()	input_pin (String)	Boolean	Compares input against stored PIN for login authentication.
to_dict()	None	Dictionary	Serializes the object into a key-value dictionary for JSON

			saving.
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3. Child Class: Admin

Description: Represents a system administrator with elevated privileges.

Inherits From: User

Attribute	Data Type	Description
<i>(Inherits All)</i>	-	Inherits user_id, name, pin, phone_number.

Method	Parameters	Return Type	Description
__init__()	user_id, name, pin, phone_number	None	Calls super().__init__().
to_dict()	None	Dictionary	Overrides parent method to inject "role": "Admin".

4. Child Class: Doctor

Description: Represents a medical professional. Contains specific attributes required for the dynamic availability algorithms.

Inherits From: User

Attribute	Data Type	Description
<i>(Inherits All)</i>	-	Inherits user_id, name, pin, phone_number.
specialization	String	Medical field (e.g., Cardiology, General Practice).
shift_start_time	String	Working hour start time in 24-hour HH:MM format.

shift_end_time	String	Working hour end time in 24-hour HH:MM format.
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Method	Parameters	Return Type	Description
<code>__init__()</code>	<i>(Base)</i> + specialization, shift_start_time, shift_end_time	None	Calls <code>super().__init__()</code> and sets new attributes.
<code>update_shifts()</code>	shift_start_time, shift_end_time	None	Modifies the doctor's working hours.
<code>to_dict()</code>	None	Dictionary	Overrides parent method to inject role, specialization, and shifts.

5. Child Class: Patient

Description: Represents a patient using the system. Holds extended demographic and contact information.

Inherits From: User

Attribute	Data Type	Description
<i>(Inherits All)</i>	-	Inherits user_id, name, pin, phone_number.
gender	String	Patient's gender.
date_of_birth	String	YYYY-MM-DD format date of birth.
email	String	Patient contact email address.
address	String	Physical residential address.

Method	Parameters	Return Type	Description
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<code>__init__()</code>	<i>(Base)</i> + gender, date_of_birth, email, address, notification=None	None	Calls <code>super().__init__()</code> and sets demographic attributes.
<code>update_profile()</code>	name, phone_number, gender, date_of_birth, email, address	Boolean	Iterates over provided updates and modifies existing attributes via <code>setattr()</code> . Returns True on completion.
<code>to_dict()</code>	None	Dictionary	Overrides parent method to inject role and demographic data.

6. Standalone Class: Appointment

Description: The transactional entity that links a Patient to a Doctor for a specific date and time block.

Attribute	Data Type	Description
<code>appointment_id</code>	String	Unique identifier (e.g., "APT-12345678").
<code>patient_id</code>	String	Foreign key linking to a Patient object.
<code>doctor_id</code>	String	Foreign key linking to a Doctor object.
<code>date</code>	String	Appointment date in YYYY-MM-DD format.
<code>start_time</code>	String	Appointment start time in HH:MM format.
<code>duration</code>	Integer	Standard appointment duration in minutes (Default: 60).
<code>status</code>	String	Lifecycle state (e.g., "Active", "Completed", "Cancelled").

Method	Parameters	Return Type	Description
<code>__init__()</code>	<code>appointment_id</code> , <code>patient_id</code> , <code>doctor_id</code> , <code>date</code> , <code>start_time</code> , <code>duration_minutes=60</code> , <code>status="Active"</code>	None	Constructor to establish the appointment relation.
<code>update_status()</code>	<code>new_status</code> (String)	None	Modifies the lifecycle state of the appointment.
<code>reschedule_appointment()</code>	<code>new_date</code> , <code>new_start_time</code>	None	Updates the temporal attributes of the appointment.
<code>to_dict()</code>	None	Dictionary	Serializes the object into a key-value dictionary for JSON saving.